2024 SSMO Team Round Solutions

SMO Team

TR 1: Find the number of ordered triples of positive integers (a, b, c) that satisfy the equation

$$2(a^b)^c + 1 = 513.$$

Answer: $\boxed{10}$

Solution: We have

$$2(a^b)^c + 1 = 513 \implies a^{bc} = 256 = 2^8.$$

So, a is a power of 2 greater than 1. Let $a = 2^x$, where x is a positive integer, so that $xbc = 8$. This means x, b, and c are all powers of 2, so let $x = 2^p$, $b = 2^q$, and $c = 2^r$, where p, q, and r are nonnegative integers with $p + q + r = 3$. From stars and bars, this equation has $\binom{5}{2} = \boxed{10}$ solutions.

TR 2: Find the sum of the three smallest positive integers n such that n^4 leaves a remainder of 1 when divided by 100.

Answer: $\boxed{51}$

Solution: From $n^4 \equiv 1 \pmod{100}$, we have $n^4 \equiv 1 \pmod 4$ and $n^4 \equiv 1 \pmod{25}$. From the first congruence, we have $n \equiv 1 \pmod 2$. Now, let $n \equiv 5a + b \pmod{25}$ for integers $0 \leq a, b < 5$. Then,

$$(5a + b)^4 = 5^4a^4 + 4 \cdot 5^3a^3b + 6 \cdot 5^2a^2b^2 + 4 \cdot 5ab^3 + b^4 \equiv 20ab^3 + b^4 \pmod{25},$$

so

$$20ab^3 + b^4 \equiv 1 \pmod{25}.$$

Now, we will proceed using casework. Clearly, $b \neq 0$, since n is not a multiple of 5. Now, note that

$$\begin{aligned} b = 1 &\implies a = 0, \\ b = 2 &\implies a = 1, \\ b = 3 &\implies a = 3, \text{ and} \\ b = 4 &\implies a = 4. \end{aligned}$$

So, we have $5a+b \in \{1, 7, 18, 24\}$. Combining this with $n \equiv 1 \pmod 2$, we find the following solutions as residues mod 50: $1, 7, 43$, and 49. So, the answer is $1 + 7 + 43 = \boxed{51}$.

TR 3: Consider four-digit positive integers N such that when N's units digit and thousands digit are removed, what remains is a two-digit perfect square. The average of all such

N can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{32743}$

Solution: Note that the two-digit perfect square obtained from removing the thousands and units digits has to be in the set $\{16, 25, \ldots, 81\}$. Since every choice of a thousands digit from 1 to 9, a square from this set, and a units digit from 0 to 9 gives a valid N, the average of N is the sum of the averages of each part. The average thousands digit is 5, the average square is $\frac{4^2+5^2+\cdots+9^2}{6} = \frac{271}{6}$, and the average units digit is $\frac{9}{2}$. So, the average of N is

$$5 \cdot 1000 + \left(\frac{271}{6}\right) \cdot 10 + \frac{9}{2} = \frac{32737}{6} \implies 32737 + 6 = \boxed{32743}.$$

TR 4: Let ABC be a right triangle with circumcenter O and incenter I such that $\angle ABC = 90^\circ$ and $\frac{AB}{BC} = \frac{3}{4}$. Let D be the projection of O onto AB, and let E be the projection of O onto BC. Let I_1 be the incenter of ADO and let I_2 be the incenter of OEC. The value of $\frac{[I_1I_2I]}{[ABC]}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{53}$

Solution: We will use analytic geometry. WLOG, let $B = (0,0)$, $A = (0,6)$, and $C = (8,0)$. Since the area of ABC is $\frac{1}{2} \cdot 8 \cdot 6 = 24$, the inradius is $\frac{2 \cdot 24}{6+8+10} = 2$, meaning $I = (2,2)$. Since ABC is a right triangle, O must be the midpoint of AC. So, D and E are the midpoints of AB and BC, respectively. Now, as ADO and OEC are both similar to ABC with ratio $\frac{1}{2}$, they have inradius 1, and we can easily compute $I_1 = (1,4)$ and $I_2 = (5,1)$. From the shoelace theorem, the area of I_1I_2I is $\frac{5}{2}$, meaning our answer is

$$\frac{\frac{5}{2}}{24} = \frac{5}{48} \implies 5 + 48 = \boxed{53}.$$

TR 5: Let ABC be a triangle with $AB = AC = 5$ and $BC = 6$. Let ω_1 be the circumcircle of ABC and let ω_2 be the circle externally tangent to ω_1 and tangent to rays AB and AC. The distance between the centers of ω_1 and ω_2 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{27}$

Solution: Let D be the midpoint of BC and let E be the point at which ω_1 and ω_2 are tangent. By symmetry, O_1 and O_2, the centers of ω_1 and ω_2, are on line AD, so E is also on line AD. This means AE is a diameter of ω_1, so $\angle ACE = 90^\circ$. Now, $AD = 4$ and $\triangle ADC \sim \triangle ACE$, so $AE = \frac{AC^2}{AD} = \frac{25}{4}$ and the radius of ω_1 is

$$r_1 = \frac{5 \cdot \frac{5}{4}}{2} = \frac{25}{8}.$$

If F is the point at which ω_2 and AC are tangent, then AO_2F is a 3-4-5 right triangle. If r_2 is the radius of ω_2, we find that

$$\frac{r_2}{r_2 + \frac{25}{4}} = \frac{3}{5},$$

so $r_2 = \frac{75}{8}$. Therefore, the final answer is

$$r_1 + r_2 = \frac{25}{2} \implies 25 + 2 = \boxed{27}.$$

TR 6: Let α, β, and γ be the roots of the polynomial $x^3 - 6x^2 - 19x - n$. If n is an integer, what is the least possible positive value of $\alpha^3 + \beta^3 + \gamma^3$?

Answer: $\boxed{3}$

Solution: From Vieta's formulas, we have

$$\alpha + \beta + \gamma = 6, \quad \alpha\beta + \alpha\gamma + \beta\gamma = -19, \quad \text{and} \quad \alpha\beta\gamma = n.$$

Since $x^3 - 6x^2 - 19x - n = 0$ for $x \in \{\alpha, \beta, \gamma\}$, we have $x^3 = 6x^2 + 19x + n$. So,

$$\begin{aligned}
\alpha^3 + \beta^3 + \gamma^3 &= (6\alpha^2 + 19\alpha + n) + (6\beta^2 + 19\beta + n) \\
&\quad + (6\gamma^2 + 19\gamma + n) \\
&= 6(\alpha^2 + \beta^2 + \gamma^2) + 19(\alpha + \beta + \gamma) + 3n \\
&= 6((\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)) \\
&\quad + 19(\alpha + \beta + \gamma) + 3n \\
&= 6((6^2) - 2(-19)) + 19(6) + 3n = 558 + 3n.
\end{aligned}$$

Since n is an integer and we are seeking to find the least positive value of $558 + 3n = 3(n + 186)$, we let $n = -185$, giving an answer of $\boxed{3}$.

TR 7: Let a and b be real numbers that satisfy

$$a^3 + 8ab^2 = 8b^3 + 4a^2b = 375.$$

Find $\lfloor ab \rfloor$.

Answer: $\boxed{12}$

Solution: Factoring the first equation, we have

$$\begin{aligned}
a^3 + 8ab^2 = 8b^3 + 4a^2b &\implies \\
a^3 - 8b^3 = 4a^2b - 8ab^2 &\implies \\
(a - 2b)(a^2 + 2ab + 4b^2) = 4ab(a - 2b) &\implies \\
(a - 2b)(a^2 - 2ab + 4b^2) = 0.
\end{aligned}$$

Consider the case where $a - 2b$ is nonzero. Then, the discriminant of the quadratic factor (in a) is $(2b)^2 - 4(4b^2) = -12b^2$. Since a and b are reals, this means $b = 0$ which is clearly not possible. So, $a = 2b$. Substituting, we have

$$8b^3 + 16b^3 = 375 \implies b^3 = \frac{125}{8} \implies b = \frac{5}{2}.$$

So,

$$\lfloor ab \rfloor = \left\lfloor (5)\left(\frac{5}{2}\right) \right\rfloor = \left\lfloor \frac{25}{2} \right\rfloor = \boxed{12}.$$

TR 8: Let $a < b < c$ be the three integers in $[0, 229)$ that satisfy the congruence $n^3 \equiv 1 \pmod{229}$. Given that $71^2 - 3$ and $107^2 + 1$ are both multiples of 229, find the value of b.

Answer: $\boxed{94}$

Solution: Note that $n^3 - 1 = (n-1)(n^2+n+1)$. Since 229 is prime, $n^3 \equiv 1 \pmod{229}$ if and only if $n \equiv 1 \pmod{229}$ or $n^2 + n + 1 \equiv 0 \pmod{229}$. Multiplying the second congruence by 4, we have $(2n+1)^2 \equiv -3 \pmod{229}$. From the given multiples of 229, we have $71^2 \equiv 3 \pmod{229}$ and $107^2 \equiv -1 \pmod{229}$, so

$$(71 \cdot 107)^2 \equiv -3 \pmod{229}.$$

Since $71 \cdot 107 = 7597 \equiv 40 \pmod{229}$, we have $(2n+1)^2 \equiv 40^2 \pmod{229}$. Since 229 is prime, this means $2n + 1 \equiv \pm 40 \pmod{229}$, so $n \equiv 134 \pmod{229}$ or $n \equiv 94 \pmod{229}$. Thus, $a = 1$, $b = \boxed{94}$, and $c = 134$.

TR 9: Let $ABCDEFGH$ be an equiangular octagon such that $AB = 6$, $BC = 8$, $CD = 10$, $DE = 12$, $EF = 6$, $FG = 8$, $GH = 10$, and $AH = 12$. The radius of the largest circle that fits inside the octagon can be expressed as $a + b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{10}$

Solution: Note that $AB \parallel EF$, $BC \parallel FG$, $CD \parallel GH$, and $DE \parallel HA$. Since opposite sides are also equal, the octagon is centrally symmetric. Any circle inside the octagon lies between each pair of opposite sides, so its diameter is at most the least distance between a pair of opposite sides. Conversely, the circle centered at the center of symmetry with this diameter fits inside the octagon, since the center is equidistant from each pair of opposite sides. Since consecutive sides meet at 135°, a side of length x at a 45° angle to a pair of opposite sides contributes $\frac{x}{\sqrt{2}}$ to the distance between them, and a side of length x perpendicular to them contributes x. So, adding the contributions of the three sides between each pair, the distances between AB and EF, BC and FG, CD and GH, and DE and HA are

$$\begin{aligned}
d_1 &= 10 + \frac{8}{\sqrt{2}} + \frac{12}{\sqrt{2}} = 10 + 10\sqrt{2}, \\
d_2 &= 12 + \frac{10}{\sqrt{2}} + \frac{6}{\sqrt{2}} = 12 + 8\sqrt{2}, \\
d_3 &= 6 + \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{2}} = 6 + 10\sqrt{2}, \text{ and} \\
d_4 &= 8 + \frac{6}{\sqrt{2}} + \frac{10}{\sqrt{2}} = 8 + 8\sqrt{2},
\end{aligned}$$

respectively. Clearly, $d_1 > d_4$ and $d_2 > d_4$, and $d_3 > d_4$ since $2\sqrt{2} > 2$. So, the least distance between a pair of opposite sides is $8 + 8\sqrt{2}$, and the radius of the largest circle that fits inside the octagon is $4 + 4\sqrt{2}$. Thus, the answer is $4 + 4 + 2 = \boxed{10}$.

TR 10: The side lengths of a convex cyclic quadrilateral $ABCD$ are integers and satisfy

$$(AB \cdot AD + BC \cdot CD)^2 = AC^2 \cdot BD^2 - 72.$$

Find the sum of all possible perimeters of $ABCD$.

Answer: $\boxed{28}$

Solution: Let $AB = a$, $BC = b$, $CD = c$, and $DA = d$. From Ptolemy's Theorem, we have $AC \cdot BD = ac + bd$. So, the equation is equivalent to

$$\begin{aligned}(ad + bc)^2 &= (ac + bd)^2 - 72 \implies \\ 72 &= a^2c^2 + b^2d^2 - a^2d^2 - b^2c^2 \implies \\ 72 &= (a^2 - b^2)(c^2 - d^2).\end{aligned}$$

Note that this equation and the perimeter do not change if we swap a with b and c with d at the same time, or if we swap the pair (a, b) with the pair (c, d). So, WLOG, assume that $a > b$, which forces $c > d$, and that $a^2 - b^2 > c^2 - d^2$, since 72 is not a perfect square. Since the difference of two perfect squares cannot be $2 \pmod 4$, we have $(a^2 - b^2, c^2 - d^2) = (72, 1), (24, 3), (9, 8)$. Clearly, $(72, 1)$ has no solutions, as $c^2 - d^2 = 1 \implies c = 1, d = 0$. For $(24, 3)$, we have $c^2 - d^2 = 3 \implies c = 2, d = 1$ and $a^2 - b^2 = 24 \implies (a - b)(a + b) = 24$. This means $(a - b, a + b) = (2, 12), (4, 6) \implies (a, b) = (7, 5), (5, 1)$. Finally, for $(9, 8)$, we have $a^2 - b^2 = 9 \implies a = 5, b = 4$, since $b \neq 0$, and $c^2 - d^2 = 8 \implies c = 3, d = 1$. This gives $(a, b, c, d) = (7, 5, 2, 1), (5, 1, 2, 1), (5, 4, 3, 1)$. It is well known that four lengths are the sides of a convex cyclic quadrilateral if and only if the longest is less than the sum of the other three. So, $(5, 1, 2, 1)$ does not work, since $5 > 1 + 2 + 1$, while $(7, 5, 2, 1)$ and $(5, 4, 3, 1)$ both work. Thus, the possible perimeters are 15 and 13, giving an answer of $15 + 13 = \boxed{28}$.

TR 11: Let S denote the set of positive divisors of 5400. Let

$$S_i = \{d \mid d \in S,\, d \equiv i \pmod 4\}$$

and let s_i denote the sum of all elements of S_i. Find the value of

$$s_0^2 + s_1^2 + s_2^2 + s_3^2 - 2s_0s_2 - 2s_1s_3.$$

Answer: $\boxed{154144400}$

Solution: Note that $5400 = 2^3 \cdot 3^3 \cdot 5^2$. Firstly, we have

$$s_0^2 + s_1^2 + s_2^2 + s_3^2 - 2s_0s_2 - 2s_1s_3 = (s_0 - s_2)^2 + (s_1 - s_3)^2.$$

Now, let $s(n)$ denote the sum of positive divisors of n. Each divisor of 5400 can be written as $2^e m$, where m is a divisor of $3^3 \cdot 5^2$, and it is $0 \pmod 4$ when $e \geq 2$ and $2 \pmod 4$ when $e = 1$. So,

$$s_0 - s_2 = (8 + 4 - 2)s(3^3 \cdot 5^2) = 10s(3^3 \cdot 5^2) = 10s(3^3)s(5^2) = 400s(5^2).$$

In addition, the odd divisor 3^i5^j is $1 \pmod 4$ when i is even and $3 \pmod 4$ when i is odd, since $3 \equiv -1 \pmod 4$ and $5 \equiv 1 \pmod 4$. So,

$$s_3 - s_1 = (27 - 9 + 3 - 1)s(5^2) = 20s(5^2).$$

It is easy to compute $s(5^2) = 31$. So, our answer is

$$(400^2 + 20^2) \cdot 31^2 = \boxed{154144400}.$$

TR 12: What is the smallest positive integer n with exactly three distinct prime factors such that for all integers k, $k^n \equiv k \pmod{n}$?

Answer: $\boxed{561}$

Solution: From Korselt's criterion, $k^n \equiv k \pmod{n}$ for all integers k if and only if n is squarefree and $p-1 \mid n-1$ for every prime $p \mid n$. This follows from Fermat's Little Theorem, along with taking $k = p$ and k a primitive root modulo p. So, $n = pqr$ for primes $p < q < r$ such that $p-1$, $q-1$, and $r-1$ all divide $n-1$. If $p = 2$, then $n-1$ is odd but $q-1$ is even, a contradiction. If $p \geq 5$, then since $5 \cdot 7 \cdot 17$ and $5 \cdot 11 \cdot 13$ both exceed 561, the only cases with $n < 561$ are $(p, q, r) = (5, 7, 11)$ and $(5, 7, 13)$. However, $11 - 1 \nmid 384$ and $5 - 1 \nmid 454$, so neither works. Finally, suppose that $p = 3$. Since $n - 1 \equiv 3q - 1 \pmod{r-1}$, we have $r - 1 \mid 3q - 1$. Since $r - 1 \geq q + 1$, we have $3q - 1 = m(r-1)$ for $m = 1$ or $m = 2$. If $m = 1$, then $r = 3q$ is not prime, so $m = 2$ and $r = \frac{3q+1}{2}$. In the same manner, $q - 1 \mid 3r - 1 = \frac{9q+1}{2}$, so $q - 1 \mid 9q + 1 = 9(q-1) + 10$, meaning $q - 1 \mid 10$. Since $q > 3$ is prime, we have $q = 11$ and $r = 17$. This gives $n = 3 \cdot 11 \cdot 17 = 561$, which works since 2, 10, and 16 all divide 560. So, our answer is $\boxed{561}$.

TR 13: In a deck of 54 cards consisting of 2 identical jokers and four identical cards of each rank $1, 2, 3, \ldots, 13$, each card is independently dealt to one of 3 people, each having a $\frac{1}{3}$ chance of receiving each card. The expected total, over the three people, of the number of distinct card types each person holds can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{917}$

Solution: From linearity of expectation, the answer is the sum, over the 14 card types, of the expected number of people holding that type. Consider the cards distinguishable for probability calculation purposes. For one of the ranks $1, 2, \ldots, 13$, there are $3^4 = 81$ equally likely ways to deal its four cards. There are 3 ways for one person to have all of them, $3! \cdot 4 + 3 \cdot \binom{4}{2} = 42$ ways for exactly two people to have them (a 3/1 or 2/2 split), and $3 \cdot \binom{4}{2} \cdot 2 = 36$ ways for all three people to have them (choose the person with two cards, those two cards, and who gets each remaining card). So, the expected number of people holding a given rank is

$$\frac{3 \cdot 1 + 42 \cdot 2 + 36 \cdot 3}{81} = \frac{65}{27}.$$

For the jokers, using a similar counting process, the expected number of people holding a joker is $\frac{3 \cdot 1 + 6 \cdot 2}{9} = \frac{5}{3}$. So, our answer is

$$\frac{5}{3} + 13 \cdot \frac{65}{27} = \frac{890}{27} \implies 890 + 27 = \boxed{917}.$$

TR 14: Let $a_1, a_2, \ldots, a_7$ be the roots of the polynomial

$$x^7 + 5x^6 + 9x^5 + x^4 + x^3 + 10x^2 + 5x + 1.$$

Find the value of

$$\left| \prod_{n=1}^{7} \prod_{m=n+1}^{7} (a_n a_m - 1) \right|.$$

Answer: $\boxed{12}$

Solution: Let $f(x)$ denote the polynomial. Each unordered pair $\{n, m\}$ with $n \neq m$ appears twice among the ordered pairs (n, m) with $1 \leq n, m \leq 7$, and the pairs with $n = m$ give the factors $a_n^2 - 1$. So,

$$\left(\prod_{n=1}^{7}\prod_{m=n+1}^{7}(a_n a_m - 1)\right)^2 \prod_{n=1}^{7}(a_n^2 - 1) = \prod_{n=1}^{7}\prod_{m=1}^{7}(a_n a_m - 1).$$

Now, note that

$$\begin{aligned}
\prod_{m=1}^{7}(a_n a_m - 1) &= -a_n^7 \prod_{m=1}^{7}\left(\frac{1}{a_n} - a_m\right) \\
&= -a_n^7 f\left(\frac{1}{a_n}\right) \\
&= -a_n^7 - (5a_n^6 + 10a_n^5 + a_n^4 + a_n^3 + 9a_n^2 + 5a_n + 1) \\
&= (5a_n^6 + 9a_n^5 + a_n^4 + a_n^3 + 10a_n^2 + 5a_n + 1) - \\
&\quad (5a_n^6 + 10a_n^5 + a_n^4 + a_n^3 + 9a_n^2 + 5a_n + 1) \\
&= -a_n^5 + a_n^2 \\
&= -(a_n)^2(a_n^3 - 1).
\end{aligned}$$

Let $\omega = e^{\frac{2\pi i}{3}}$, so that $x^3 - 1 = (x-1)(x-\omega)(x-\omega^2)$. So,

$$\begin{aligned}
\prod_{n=1}^{7}\prod_{m=1}^{7}(a_n a_m - 1) &= \prod_{n=1}^{7}\left(-(a_n)^2(a_n^3 - 1)\right) \\
&= -\left(\prod_{n=1}^{7} a_n\right)^2 \prod_{n=1}^{7}(a_n - 1) \\
&\quad \cdot \prod_{n=1}^{7}(a_n - \omega)\prod_{n=1}^{7}(a_n - \omega^2) \\
&= -(-1)^2(-f(1))\left(-f(\omega)\right)\left(-f(\omega^2)\right) \\
&= f(1)f(\omega)f(\omega^2).
\end{aligned}$$

Since $\omega^3 = 1$ and $\omega^2 + \omega + 1 = 0$, we have $f(\omega) = 19\omega^2 + 7\omega + 7 = 12\omega^2$ and $f(\omega^2) = 12\omega^4 = 12\omega$. So,

$$f(\omega)f(\omega^2) = 144\omega^3 = 144.$$

Thus,

$$\prod_{n=1}^{7}\prod_{m=1}^{7}(a_n a_m - 1) = f(1)f(\omega)f(\omega^2) = 33 \cdot 144.$$

Now, we have

$$\begin{aligned}
\prod_{n=1}^{7}(a_n^2 - 1) &= \prod_{n=1}^{7}(a_n - 1)\prod_{n=1}^{7}(a_n + 1) \\
&= (-f(1))(-f(-1)) \\
&= f(1)f(-1) \\
&= 33 \cdot 1 = 33.
\end{aligned}$$

Substituting this into the first equation, we have

$$\left(\prod_{n=1}^{7}\prod_{m=n+1}^{7}(a_na_m-1)\right)^2=\frac{33\cdot 144}{33}=144\implies\left|\prod_{n=1}^{7}\prod_{m=n+1}^{7}(a_na_m-1)\right|=\sqrt{144}=\boxed{12}.$$

TR 15: In triangle ABC inscribed in circle ω, let M be the midpoint of BC. Denote P as the second intersection of line AM with ω, where $P\neq A$. If $BP=9$, $CP=13$, and $AM=20$, find the perimeter of triangle ABC.

Answer: $\boxed{64}$

Solution: We will use barycentric coordinates with ABC as the reference triangle. Let $A=(1:0:0)$, $B=(0:1:0)$, and $C=(0:0:1)$. Note that the circumcircle of ABC can be represented as $a^2yz+b^2xz+c^2xy=0$. Since P lies on cevian AM, with $M=\left(0:\frac{1}{2}:\frac{1}{2}\right)$, we have $P=(x_p:y_p:z_p)$, with $y_p=z_p$. Substituting into the equation for the circumcircle of ABC, we have

$$\begin{aligned}a^2y_pz_p+b^2x_pz_p+c^2x_py_p=0&\implies\\ a^2y_p^2+b^2x_py_p+c^2x_py_p=0&\implies\\ a^2y_p=-(b^2+c^2)x_p&\implies\\ y_p=-\frac{b^2+c^2}{a^2}x_p.&\end{aligned}$$

From $x_p+y_p+z_p=1$, we have

$$x_p+\left(-\frac{b^2+c^2}{a^2}\right)x_p+\left(-\frac{b^2+c^2}{a^2}\right)x_p=1\implies x_p=-\frac{a^2}{2b^2+2c^2-a^2}.$$

So,

$$P=\left(-\frac{a^2}{2b^2+2c^2-a^2}:\frac{b^2+c^2}{2b^2+2c^2-a^2}:\frac{b^2+c^2}{2b^2+2c^2-a^2}\right).$$

Now, we have

$$\overrightarrow{PB}=(x_{pb},y_{pb},z_{pb})=\left(\frac{a^2}{2b^2+2c^2-a^2},\frac{b^2+c^2-a^2}{2b^2+2c^2-a^2},-\frac{b^2+c^2}{2b^2+2c^2-a^2}\right).$$

So,

$$\begin{aligned}
PB^2 &= \left|a^2 y_{pb} z_{pb} + b^2 x_{pb} z_{pb} + c^2 x_{pb} y_{pb}\right| \\
&= \left|a^2 \left(\frac{b^2+c^2-a^2}{2b^2+2c^2-a^2}\right)\left(-\frac{b^2+c^2}{2b^2+2c^2-a^2}\right)\right. \\
&\quad + b^2 \left(\frac{a^2}{2b^2+2c^2-a^2}\right)\left(-\frac{b^2+c^2}{2b^2+2c^2-a^2}\right) \\
&\quad \left. + c^2 \left(\frac{a^2}{2b^2+2c^2-a^2}\right)\left(\frac{b^2+c^2-a^2}{2b^2+2c^2-a^2}\right)\right| \\
&= \frac{|a^4b^2 - 2a^2b^4 - 2a^2b^2c^2|}{(2b^2+2c^2-a^2)^2} \\
&= \frac{a^2b^2(2b^2+2c^2-a^2)}{(2b^2+2c^2-a^2)^2} \\
&= \frac{a^2b^2}{2b^2+2c^2-a^2} \implies \\
PB &= \frac{ab}{\sqrt{2b^2+2c^2-a^2}}.
\end{aligned}$$

In the same manner, we have

$$PC = \frac{ac}{\sqrt{2b^2+2c^2-a^2}}.$$

From the median length formula, we have $AM = \frac{1}{2}\sqrt{2b^2+2c^2-a^2}$, so

$$AM = 20 \implies \frac{\sqrt{2b^2+2c^2-a^2}}{2} = 20 \implies \sqrt{2b^2+2c^2-a^2} = 40.$$

This means

$$ab = PB\sqrt{2b^2+2c^2-a^2} = 9 \cdot 40$$

and

$$ac = PC\sqrt{2b^2+2c^2-a^2} = 13 \cdot 40.$$

Solving for b and c, we have $b = \frac{9\cdot 40}{a}, c = \frac{13\cdot 40}{a}$. Substituting, we get

$$\begin{aligned}
\sqrt{2b^2+2c^2-a^2} &= 40 \implies \\
2b^2+2c^2-a^2 &= 40^2 \implies \\
2\left(\left(\frac{9\cdot 40}{a}\right)^2 + \left(\frac{13\cdot 40}{a}\right)^2\right) - a^2 &= 40^2 \implies \\
a^4 + 40^2a^2 - 2\left((9\cdot 40)^2 + (13\cdot 40)^2\right) &= 0 \implies \\
a^2 &= \frac{-40^2 \pm \sqrt{40^4 + 8\left((9\cdot 40)^2 + (13\cdot 40)^2\right)}}{2} \\
&= (20)\left(\pm\sqrt{40^2 + 8(9^2+13^2)} - 40\right) \\
&= (20)\left(\pm 60 - 40\right) = 400, -2000 \implies \\
a &= 20.
\end{aligned}$$

So, $b = \frac{9\cdot 40}{a} = \frac{9\cdot 40}{20} = 18$ and $c = \frac{13\cdot 40}{20} = 26$, giving a perimeter of

$$20 + 18 + 26 = \boxed{64}.$$